

Production Trends and Asset Concentration Analysis in Midland County Using Texas Railroad Commission Public Data

SPE-Style Technical Report

Midland County, Texas | January 2020 - March 2026

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Metric	Result
Total oil analyzed	1,379,471,252 BBL
Total gas analyzed	4,860,365,156 MCF
Oil growth, 2020-2025	26.1%
Gas growth, 2020-2025	78.7%
Top field	SPRABERRY (TREND AREA)
Top operator	PIONEER NATURAL RES. USA, INC.
Leases analyzed	5,674
Top 10% lease oil share	58.1%

Abstract

Midland County, Texas is one of the most active hydrocarbon-producing areas in the Midland Basin portion of the Permian Basin. This study analyzes public production records obtained from the Texas Railroad Commission Production Data Query system for the period January 2020 through March 2026. The analysis evaluates production at the county, field, operator, and lease levels using five RRC CSV exports: District, Field, Lease, Monthly Totals, and Operator reports.

The study found that Midland County oil production increased 26.1% from 2020 to 2025, while gas production increased 78.7% during the same period. The highest-producing field was SPRABERRY (TREND AREA), and the highest-producing operator was PIONEER NATURAL RES. USA, INC.. At the lease level, 5,674 leases were analyzed after removing the RRC total row. The top 10% of leases accounted for approximately 58.1% of lease-level oil production, demonstrating a high degree of production concentration among a relatively small number of assets.

The results show that publicly available regulatory data can be used to generate practical production insights relevant to petroleum engineering, production analysis, and asset benchmarking. The workflow also demonstrates a reproducible approach for cleaning raw production records, developing summary tables, and producing technical figures for communication of results.

1. Introduction

The Permian Basin is one of the most important oil and gas producing regions in the United States. Within the broader basin, Midland County is located in the Midland Basin and includes major production from the Spraberry and related producing intervals. Production analysis in this area is useful because it provides insight into regional output trends, operator activity, field dominance, and concentration of production across lease-level assets.

This project was designed as an independent technical analysis using only public data. Instead of relying on commercial databases, the study uses Texas Railroad Commission production records, which are publicly accessible and commonly used to verify oil and gas activity in Texas. The goal was to transform raw regulatory data into a structured engineering-style report that can support discussion of production behavior, data analysis methods, and future research opportunities.

The analysis focuses on Midland County only. This narrow scope makes the project manageable while still providing enough data for meaningful results. The workflow can later be extended to additional Permian Basin counties such as Martin, Reeves, Loving, Howard, or Upton for a broader county-benchmarking study.

2. Objectives

The main objectives of this study were to quantify production trends in Midland County, identify leading operators and fields, evaluate gas-oil ratio behavior, and measure production concentration across leases. The project also aimed to create a reproducible workflow that converts raw Texas Railroad Commission exports into cleaned tables, figures, and a technical report.

The specific questions addressed were: how did Midland County oil and gas production change from 2020 to 2025; which operators and fields contributed the most oil production; how has the county-level gas-oil ratio changed over time; and what share of oil production came from the highest-producing 10% of leases?

3. Data Source and Scope

Data were obtained from the Texas Railroad Commission Production Data Query system. Five CSV exports were used: District Report, Field Report, Lease Report, Monthly Totals Report, and Operator Report. Together, these files provided county-level monthly production, field totals, operator totals, and lease-level production records for Midland County.

The analysis period covered January 2020 through March 2026. The dataset included 75 monthly observations after cleaning, 94 field-level records, 186 operator rows before final ranking, and 5,674 lease-level records after removing the RRC total row. Total analyzed production across the cleaned field-level dataset was 1,379,471,252 BBL of oil and 4,860,365,156 MCF of gas.

Table 1 - Annual Midland County Production Summary

Year	Months in Dataset	Oil (BBL)	Gas Total (MCF)	GOR (MCF/BBL)
2020.0	12.0	191,675,933	549,802,603	2.87
2021.0	12.0	199,143,122	630,968,700	3.17
2022.0	12.0	221,678,081	720,013,945	3.25
2023.0	12.0	234,518,111	829,080,596	3.54
2024.0	12.0	234,762,267	915,274,251	3.90
2025.0	12.0	241,692,085	982,604,254	4.07
2026.0	3.0	56,001,653	232,620,807	4.15

4. Methodology

The workflow consisted of data loading, cleaning, aggregation, ranking, concentration analysis, and visualization. The raw CSV exports included report header information above the actual data table, so the analysis skipped the report metadata rows before loading the tabular data. Production columns were converted from text to numeric values, commas were removed, and blank or nonnumeric entries were handled as zeros where appropriate.

Oil production was measured in barrels (BBL). Total gas production was calculated as casinghead gas plus gas-well gas and reported in thousand cubic feet (MCF). Gas-oil ratio (GOR) was calculated as total gas divided by oil production, expressed in MCF/BBL. Operator, field, and lease tables were ranked by cumulative oil production.

For the lease concentration analysis, leases were sorted from highest to lowest cumulative oil production. Cumulative oil share was then calculated relative to total lease-level oil production. The top 10% threshold included 568 leases out of 5,674 analyzed leases. This method measures how much oil output is concentrated in the highest-producing lease assets.

5. Results

5.1 County Oil Production Trend

Midland County oil production increased from 191,675,933 BBL in 2020 to 241,692,085 BBL in 2025. This represents a 26.1% increase over the five-year comparison period. Monthly oil volumes show a general upward trend after 2020, although the data include visible month-to-month volatility and notable dips around early 2021 and early 2026.

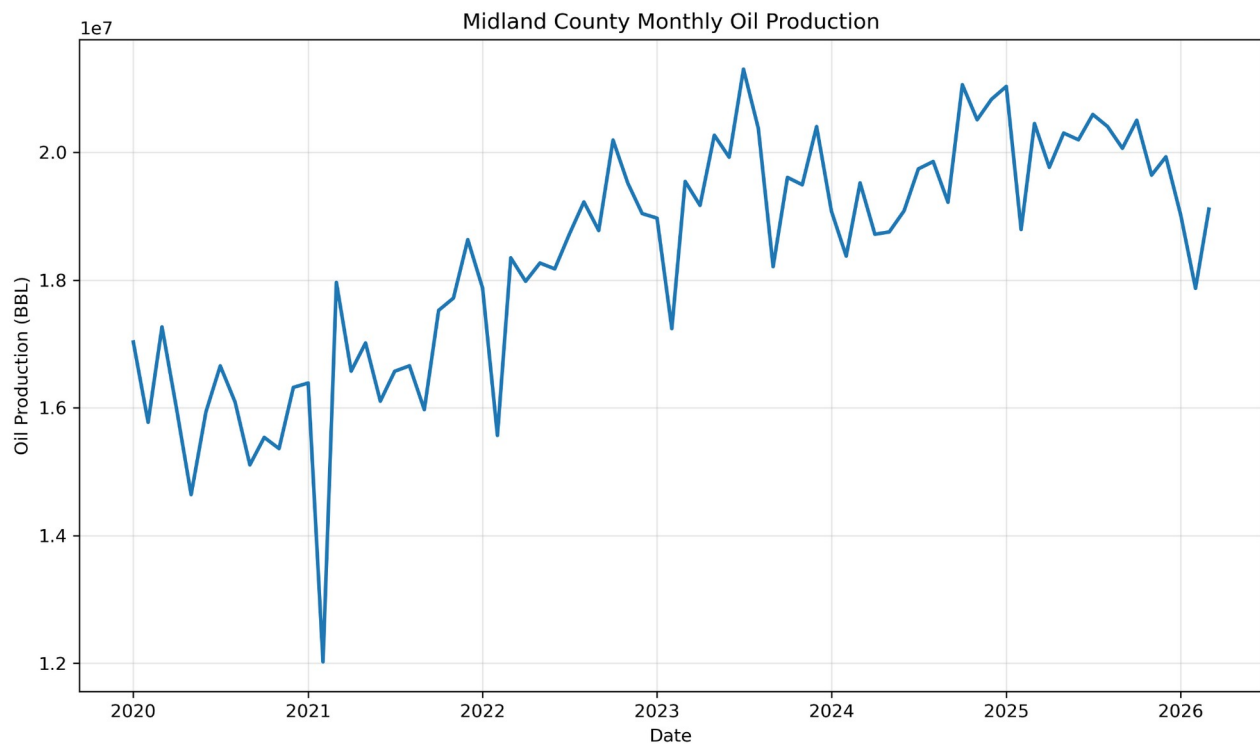


Figure 1 - Midland County monthly oil production from January 2020 through March 2026.

5.2 County Gas Production Trend

Gas production increased more rapidly than oil production over the same period. Midland County gas production increased from 549,802,603 MCF in 2020 to 982,604,254 MCF in 2025, representing a 78.7% increase. The stronger increase in gas production suggests that the county became increasingly gas-rich in its reported production mix during the study period.

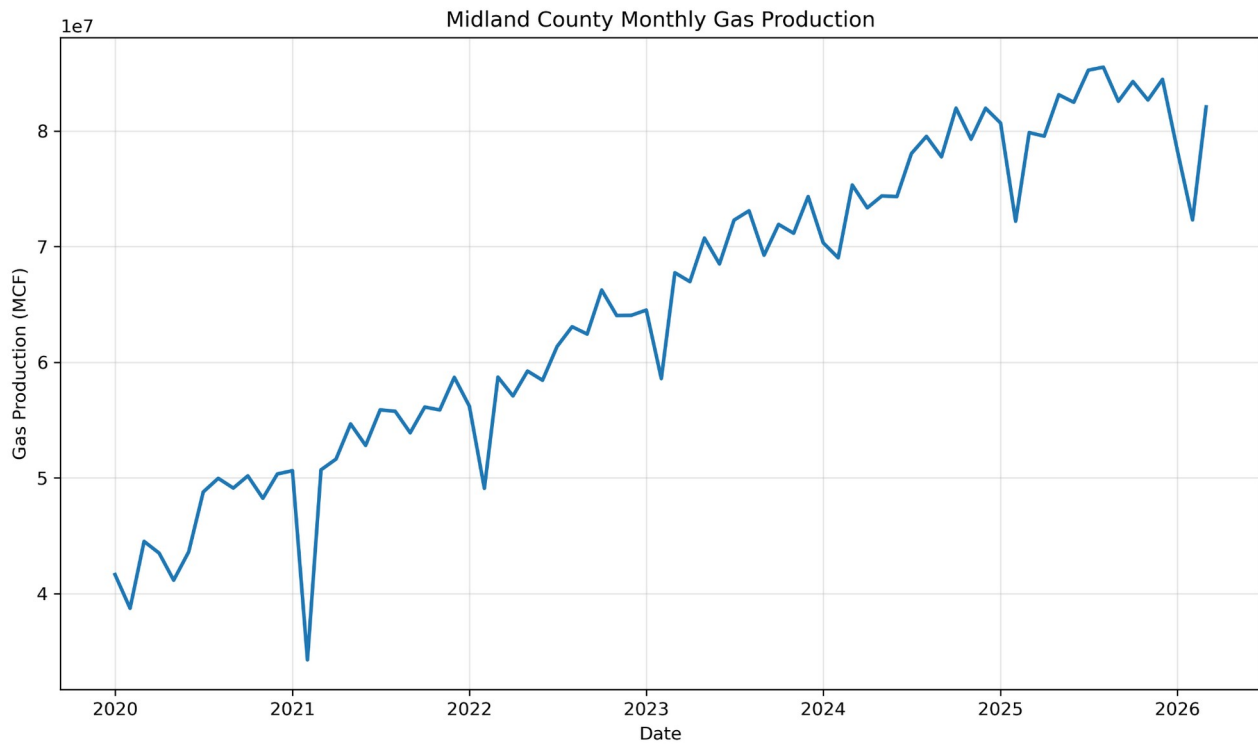


Figure 2 - Midland County monthly gas production from January 2020 through March 2026.

5.3 Gas-Oil Ratio Trend

The county-level GOR increased from an annual average of 2.87 MCF/BBL in 2020 to 4.07 MCF/BBL in 2025. The overall dataset average GOR was 3.52 MCF/BBL. This trend is consistent with gas production growing faster than oil production. The analysis does not isolate the specific cause of the increase because the RRC production exports do not include completion design, reservoir interval, lateral length, or well vintage data.

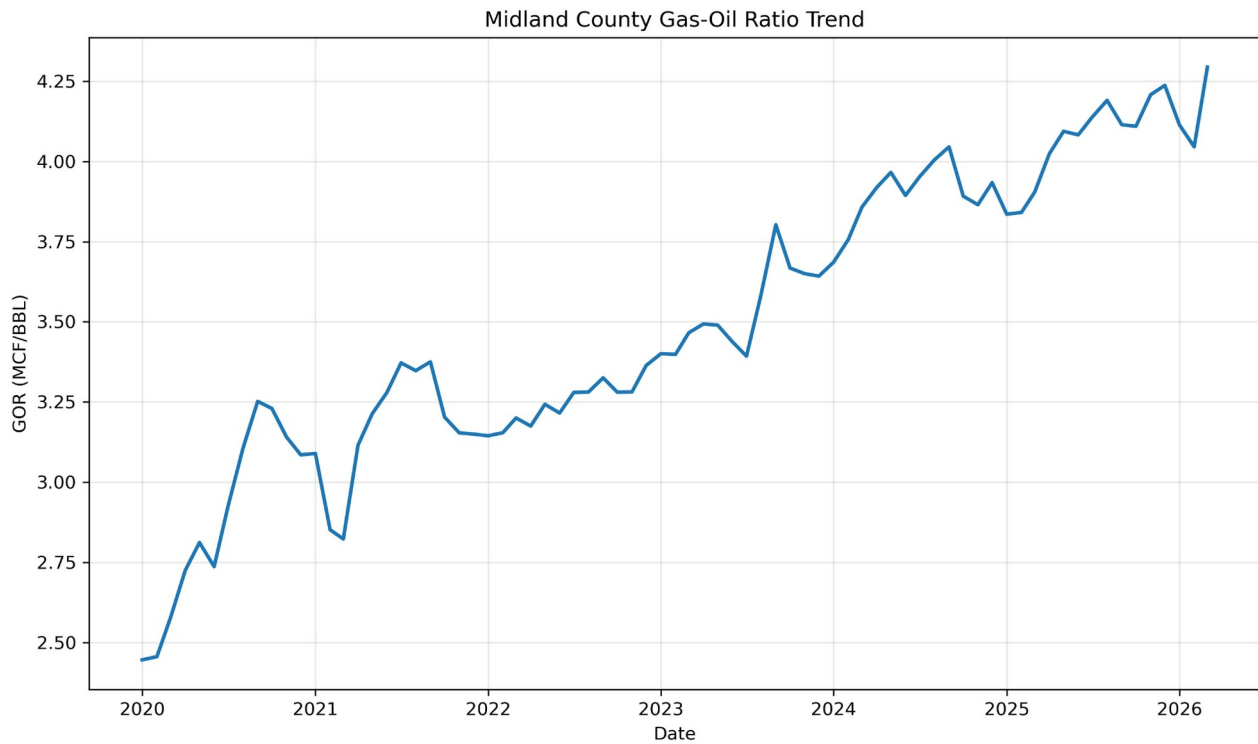


Figure 3 - Midland County gas-oil ratio trend, calculated as total gas production divided by oil production.

5.4 Operator Ranking

Operator-level analysis showed that PIONEER NATURAL RES. USA, INC. was the leading operator by cumulative oil production. The top ten operators include several large Permian producers, including Pioneer Natural Resources USA, Endeavor Energy Resources, XTO Energy, Chevron, Diamondback, COG Operating, Apache, Ovintiv, CrownQuest, and Permian Deep Rock Oil. The top operator alone represented approximately 33.5% of field-level county oil production in the cleaned operator summary.

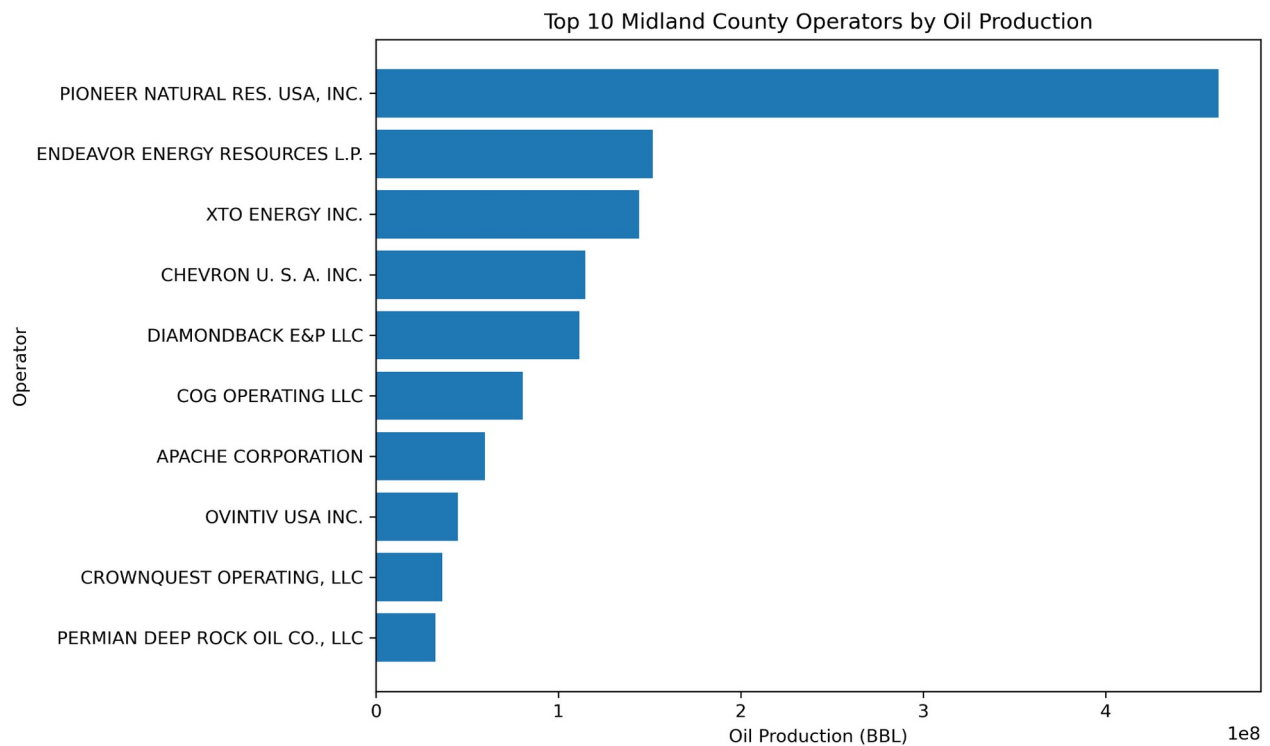


Figure 4 - Top 10 Midland County operators by cumulative oil production.

Table 2 - Top 10 Operators by Oil Production

Rank	Operator Name	Oil (BBL)	Gas Total (MCF)	GOR (MCF/BBL)	Share of County Oil
1	PIONEER NATURAL RES. USA, INC.	461,756,167	1,573,697,233	3.41	33.5%
2	ENDEAVOR ENERGY RESOURCES L.P.	151,813,457	437,944,654	2.88	11.0%
3	XTO ENERGY INC.	144,255,030	698,852,985	4.84	10.5%
4	CHEVRON U. S. A. INC.	114,741,932	359,035,478	3.13	8.3%
5	DIAMONDBACK E&P LLC	111,512,115	469,461,067	4.21	8.1%
6	COG OPERATING LLC	80,378,990	270,902,918	3.37	5.8%
7	APACHE CORPORATION	59,634,834	186,858,394	3.13	4.3%
8	OVINTIV USA INC.	45,012,707	235,748,375	5.24	3.3%
9	CROWNQUEST OPERATING, LLC	36,465,637	139,906,375	3.84	2.6%
10	PERMIAN DEEP ROCK OIL CO., LLC	32,774,650	70,023,183	2.14	2.4%

5.5 Field Ranking

Field-level production was dominated by SPRABERRY (TREND AREA). This field accounted for approximately 81.1% of the field-level oil production analyzed. The second-ranked field was Spraberry (Trend Area) R 40 EXC, followed by Parks (Consolidated). The dominance of the Spraberry trend is expected because the Spraberry has historically been one of the key producing systems in the Midland Basin.

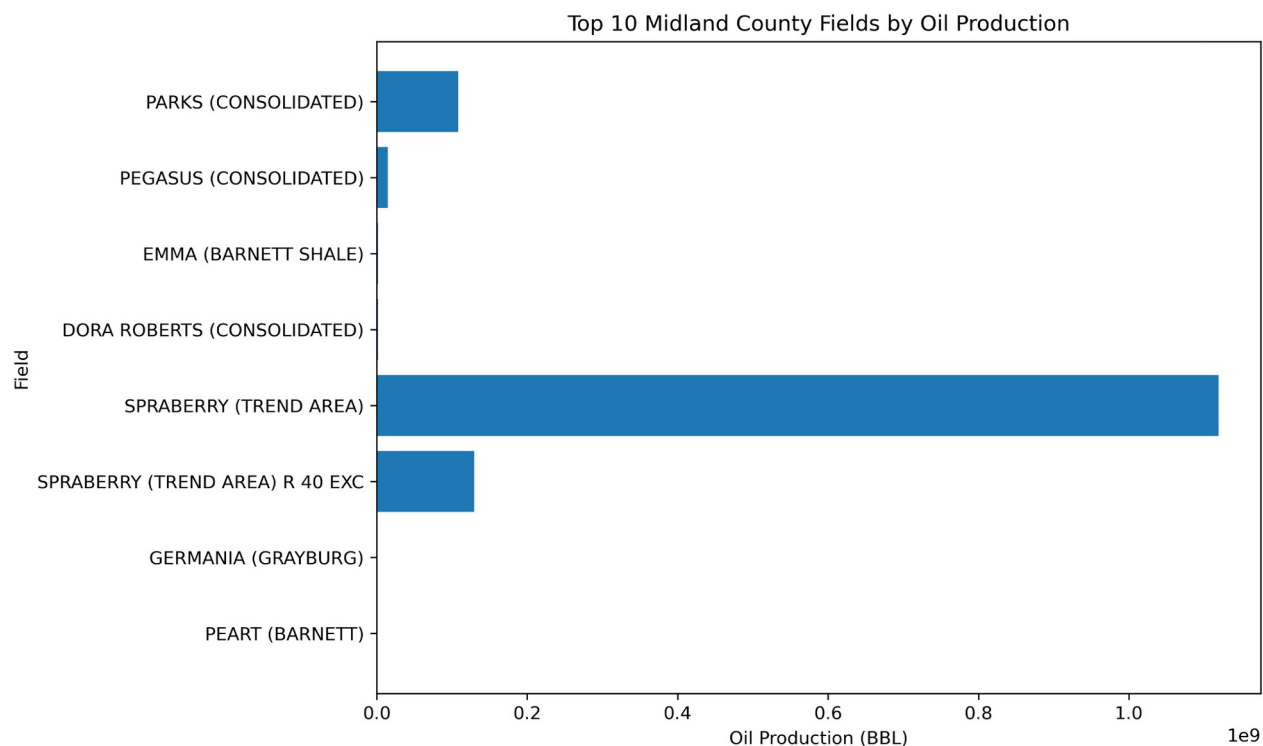


Figure 5 - Top 10 Midland County fields by cumulative oil production.

Table 3 - Top 10 Fields by Oil Production

Rank	Field Name	Oil (BBL)	Gas Total (MCF)	GOR (MCF/BBL)	Share of County Oil
1	SPRABERRY (TREND AREA)	1,119,241,146	3,628,915,534	3.24	81.1%
2	SPRABERRY (TREND AREA) R 40 EXC	129,676,142	787,274,092	6.07	9.4%
3	PARKS (CONSOLIDATED)	108,221,412	294,759,617	2.72	7.8%
4	PEGASUS (CONSOLIDATED)	14,920,092	87,644,426	5.87	1.1%
5	EMMA (BARNETT SHALE)	1,562,968	11,774,484	7.53	0.1%
6	DORA ROBERTS (CONSOLIDATED)	1,524,912	9,569,679	6.28	0.1%
7	SPRABERRY (TREND AREA)	1,425,332	3,369,559	2.36	0.1%
8	SPRABERRY (TREND AREA) R 40 EXC	703,093	12,792,270	18.19	0.1%
9	GERMANIA (GRAYBURG)	275,137	33,697	0.12	0.0%
10	PEAT (BARNETT)	267,373	1,986,730	7.43	0.0%

5.6 Lease-Level Concentration

Lease-level analysis provides the most granular view in this study. The RRC lease export contained 5,675 rows before cleaning. After removing the RRC total row, 5,674 leases were analyzed. The top 10% of leases, equal to 568 leases, accounted for approximately 58.1% of lease-level oil production. This indicates that production is highly concentrated among a relatively small subset of assets.

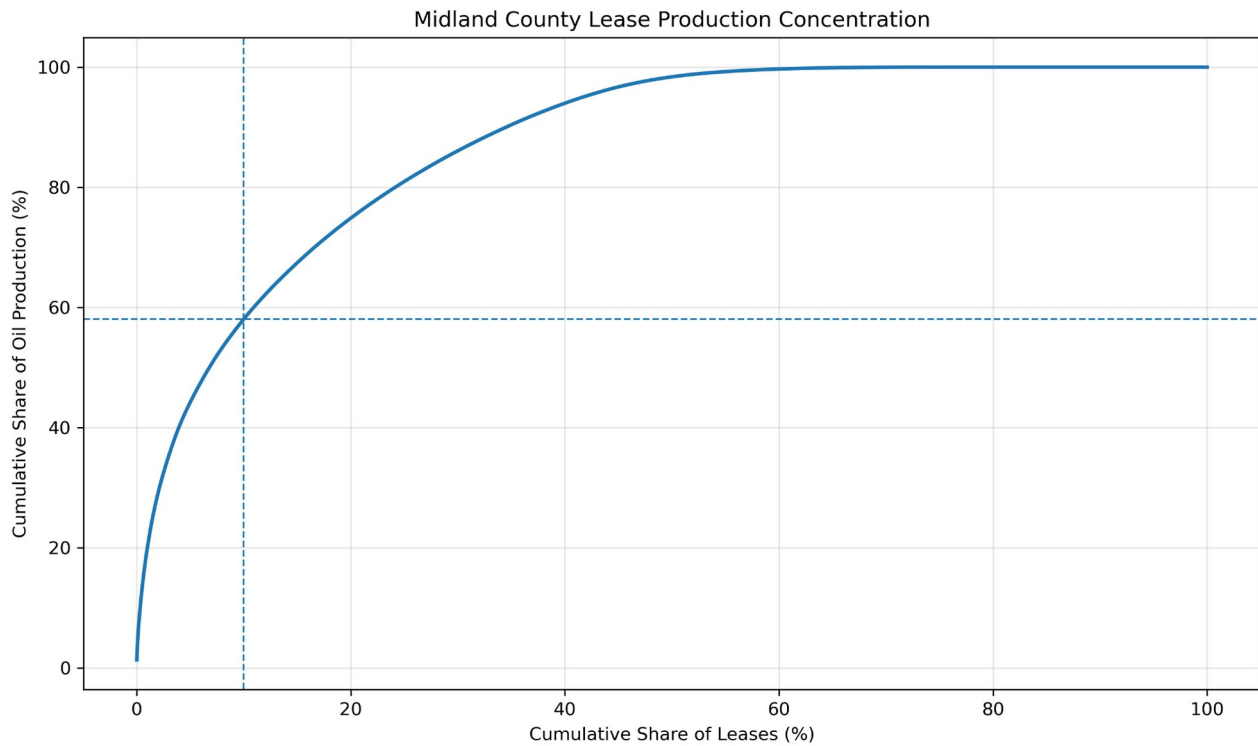


Figure 6 - Lease production concentration curve showing cumulative share of oil production by cumulative share of leases.

Table 4 - Top 10 Leases by Oil Production

Rank	Lease Name	Oil (BBL)	Gas Total (MCF)	GOR (MCF/BBL)	Share of County Oil
1	TEXAS TEN Y PU	18,773,680	79,591,550	4.24	1.4%
2	SOA SCHARBAUER SE	13,745,848	41,746,441	3.04	1.0%
3	SCHARBAUER, C.	11,007,423	29,688,179	2.70	0.8%
4	NOBLES	8,536,846	17,812,571	2.09	0.6%
5	SOUTH CURTIS RANCH	8,223,656	15,634,889	1.90	0.6%
6	PEGASUS FIELD UNIT 3	7,595,628	51,123,303	6.73	0.6%
7	BRD ASTRODOME UNIT	7,453,582	27,339,206	3.67	0.5%
8	GBG REEDY 13/25 UNIT	7,135,703	25,240,577	3.54	0.5%
9	HUTT, DONALD L.'C'	6,668,492	35,217,798	5.28	0.5%
10	GRIDIRON	5,909,076	20,968,411	3.55	0.4%

6. Discussion

The results show three major trends. First, Midland County remained a strong oil producer over the study period, with oil production increasing by more than one quarter from 2020 to 2025. Second, gas production increased much faster than oil production, producing a visible upward trend in GOR. Third, production is not evenly distributed across all leases or fields; a small percentage of producing assets contribute a majority of lease-level oil production.

The rising GOR is one of the most important technical observations. Because the RRC production exports are aggregated and do not include completion or reservoir details, the analysis should not claim a single cause. Potential explanations could include a changing mix of producing intervals, maturing wells, new development in gassier areas, operational changes, or differences in reporting composition. A follow-up project could link production to well-level completion data and well vintage to test these explanations more rigorously.

The lease concentration finding also has practical significance. If the top 10% of leases account for approximately 58.1% of oil production, then operator performance, acreage quality, facility constraints, and development strategy likely have a major effect on county-level output. From an engineering standpoint, this type of concentration analysis can help identify which assets deserve deeper decline-curve, spacing, or completion analysis.

7. Limitations

This study uses aggregated public regulatory data, which creates several limitations. The RRC data used here do not include lateral length, proppant loading, stage count, completion date, true measured depth, reservoir quality, pressure data, or operating cost. As a result, the study can identify production patterns but cannot fully explain the engineering causes behind those patterns.

The analysis also does not perform well-level decline-curve analysis. Lease-level records are useful for identifying production concentration, but a more advanced reservoir or production engineering study would require individual well identifiers, first production dates, completion timing, and well-level monthly production histories. The results should therefore be interpreted as a county-level production benchmarking study rather than a complete reservoir engineering evaluation.

8. Future Work

The most valuable next step would be to expand the workflow to additional Permian Basin counties. Comparing Midland County against Martin, Reeves, Loving, Howard, and Upton Counties would allow a broader analysis of county-level production growth, operator positioning, GOR trends, and asset concentration across the Permian Basin.

Additional future work could include decline-curve analysis, operator-normalized performance metrics, mapping of lease or well locations, integration of drilling permit data, and a comparison of production trends by field or formation. A machine-learning extension could attempt to predict first-year production using county, operator, field, vintage, and available well characteristics if suitable well-level data are obtained.

9. Conclusions

- Midland County oil production increased 26.1% from 2020 to 2025.
- Midland County gas production increased 78.7% over the same period, indicating that gas grew much faster than oil.
- The annual GOR increased from 2.87 MCF/BBL in 2020 to 4.07 MCF/BBL in 2025.
- PIONEER NATURAL RES. USA, INC. was the top operator by cumulative oil production.
- SPRABERRY (TREND AREA) was the top field by cumulative oil production.
- After removing the RRC total row, 5,674 leases were analyzed.

- The top 10% of leases accounted for approximately 58.1% of lease-level oil production.

References

Texas Railroad Commission. Production Data Query System. Midland County District, Field, Lease, Monthly Totals, and Operator CSV exports. Data downloaded and analyzed for January 2020 through March 2026.

Appendix A - Reproducibility Notes

The cleaned workbook and figures were recreated from five raw Texas Railroad Commission CSV exports using a Python workflow. The workflow loads the raw files, skips RRC metadata rows, cleans column names, converts production values to numeric fields, calculates total gas and GOR, ranks operators, fields, and leases by oil production, performs lease concentration analysis, exports a cleaned Excel workbook, and saves the figures used in this report.

Raw file names used in the workflow: ProductionDataQuery_District_ReportCsv.csv;
ProductionDataQuery_Field_ReportCsv.csv; ProductionDataQuery_Lease_ReportCsv.csv;
ProductionDataQuery_monthlyTotals_ReportCsv.csv; ProductionDataQuery_Operator_ReportCsv.csv.